# Lesson 3 Teacher's Key

GameMaker

**INSTRUCTOR ONLY - do not distribute to students**

## Level 1 - Complete Worked Solution

obj_bullet's own Step event, plus an addition to obj_player's existing Step event:

```
// obj_bullet's Step event:
y -= 8;
if (y < 0) instance_destroy();

// Add to obj_player's existing Step event:
if (keyboard_check_pressed(vk_space)) {
    instance_create_layer(x, y, "Instances", obj_bullet);
}
```

## Level 2 - Complete Worked Solution

keyboard_check_pressed() is already edge-triggered (true for exactly one frame per physical press), so Level 1's fire check needs no changes for Level 2. An optional tunable cooldown, if a slower fire rate is wanted (add fire_cooldown = 0; to obj_player's Create event):

```
// obj_player's Step event:
if (fire_cooldown > 0) fire_cooldown -= 1;
if (keyboard_check_pressed(vk_space) && fire_cooldown <= 0) {
    instance_create_layer(x, y, "Instances", obj_bullet);
    fire_cooldown = 12;
}
```

## Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

### Sound: play on fire

Add sfx_shoot.wav as a sound resource (keep its default name, sfx_shoot):

```
if (keyboard_check_pressed(vk_space) && fire_cooldown <= 0) {
    instance_create_layer(x, y, "Instances", obj_bullet);
    fire_cooldown = 12;
    audio_play_sound(sfx_shoot, 1, false);
}
```

### Visual: give the bullet its own look

First, in the Sprite Editor, set spr_bullet's Origin to Center. image_xscale/image_yscale scale the sprite around its Origin point, not around its visual center - with GameMaker's default top-left Origin, scaling up makes the bullet visibly grow to the right and down (a real position shift, not just a size change) instead of scaling evenly in place. This is a one-time sprite setting, not code.

The flash itself needs to be temporary, so it needs a countdown timer to reset it - the same fire_cooldown-style pattern from Level 2, just living on obj_bullet instead of obj_player. Set the flashy values plus a timer in Create, then count the timer down in Step and reset back to normal the instant it runs out:

```
// obj_bullet's Create event:
image_blend = c_yellow;
image_xscale = 1.3;
image_yscale = 1.3;
flash_timer = 6; // about 0.1 seconds at 60fps

// obj_bullet's Step event - add alongside the existing movement code:
if (flash_timer > 0) {
    flash_timer -= 1;
    if (flash_timer <= 0) {
        image_blend = c_white;
        image_xscale = 1;
        image_yscale = 1;
    }
}
```

### Structure: keep bullet behavior in obj_bullet

obj_player's Step event should only ever trigger a shot (instance_create_layer, cooldown, sound) - everything about what a bullet actually does once it exists (movement, off-screen cleanup, its own look) belongs in obj_bullet's own events. Keeping this boundary clean is what let the Sound and Visual ideas above each drop into one object without touching the other.

### Complete Level 3 solution (all 3 goals combined)

Everything above combined, across obj_bullet and obj_player:

```
// spr_bullet's Origin (Sprite Editor, one-time setting): Center

// obj_bullet's Create event:
image_blend = c_yellow;
image_xscale = 1.3;
image_yscale = 1.3;
flash_timer = 6; // about 0.1 seconds at 60fps

// obj_bullet's Step event:
y -= 8;
if (y < 0) instance_destroy();
if (flash_timer > 0) {
    flash_timer -= 1;
    if (flash_timer <= 0) {
        image_blend = c_white;
        image_xscale = 1;
        image_yscale = 1;
    }
}
```

```
// obj_player's Create event:
fire_cooldown = 0;

// obj_player's Step event:
if (fire_cooldown > 0) fire_cooldown -= 1;
if (keyboard_check_pressed(vk_space) && fire_cooldown <= 0) {
    instance_create_layer(x, y, "Instances", obj_bullet);
    fire_cooldown = 12;
    audio_play_sound(sfx_shoot, 1, false);
}
```

## Alternate Path Notes: GML Visual

"If Key Pressed" is already edge-triggered the same as keyboard_check_pressed(), so the same no-extra-cooldown-needed-for-Level-2 logic applies in blocks too.

Movement uses "Jump To Point" (Movement category) with Relative checked on BOTH X and Y, the same purpose-built block from Level 1, rather than the generic Assign Variable/Set Variable - watch for students who only check Relative on the axis that's changing (Y) and leave X unchecked, which silently snaps X to an absolute 0 every step. "Instance - Create Instance" needs the same both-axes Relative treatment when spawning obj_bullet at the player's position.

The Level 3 goals map onto blocks the same way they map onto code, favoring the most specific named block available: "Play Audio" (Audio category) for sound, and "Set Instance Variable" for image_blend/image_xscale/image_yscale/flash_timer - all recognized built-ins on Set Instance Variable's own dropdown (Image Blend, Horizontal Scale, Vertical Scale), so there's no need for the generic Assign Variable here either. The Visual idea's flash needs spr_bullet's Origin set to Center first (Sprite Editor) - image_xscale/image_yscale scale around the Origin, and GameMaker's default top-left Origin makes the bullet visibly shift right and down as it scales instead of growing in place - and needs an "If Variable"-gated countdown on flash_timer in Step to actually reset image_blend/image_xscale/image_yscale back to normal, the same shape as the code path's timer. All of this stays inside obj_bullet's own events, kept separate from obj_player's events the same way the code path keeps the two objects separate.

- Keeping the same Structure principle (bullet behavior lives with the bullet, not the player) whether a student is in Code or Visual reinforces that it's a design idea, not a syntax rule.